

Floyd Cycle Detection Algorithm

- Linked list or nums in $[1, n]$

Algorithm!

① slow pointer moves 1x speed

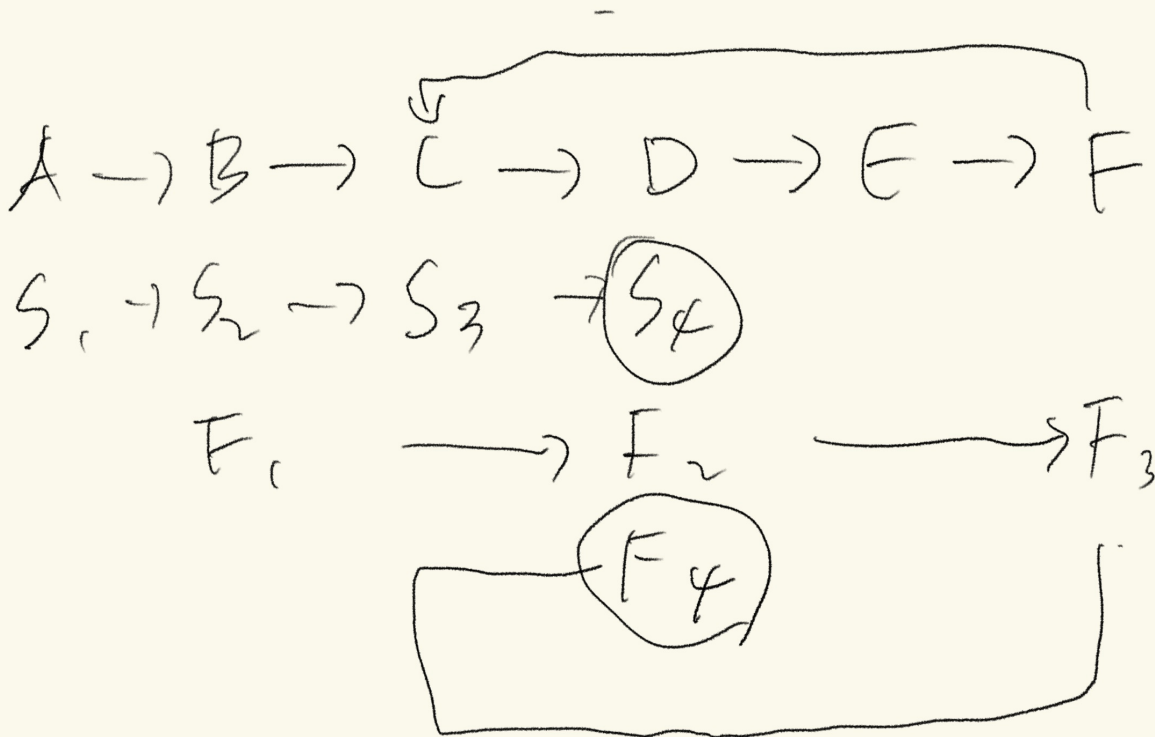
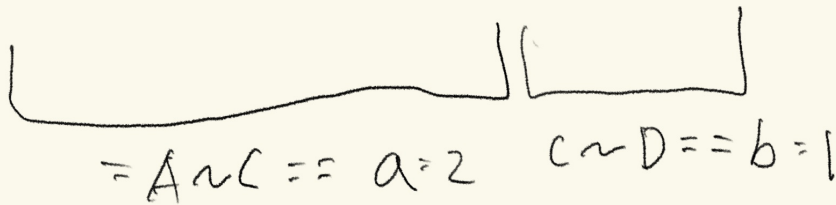
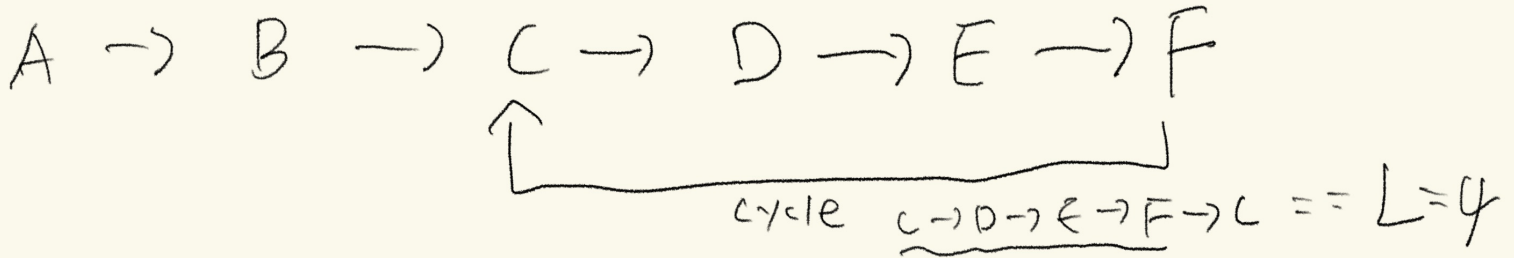
② fast pointer moves 2x speed

⇒ They meet in somewhere of cycle

① move slow to the beginning

⇒ ② move the both pointers 1x speed

⇒ They meet at the entrance of cycle



cycle entrance = C

distance from start to cycle entrance = D

$$a: A \sim C \Rightarrow 2$$

$$b: C \sim D \Rightarrow 1$$

$$L: C \sim F \Rightarrow 4$$

slow pointer 이동할 때 : $A \sim D$

$$\therefore \text{slow} = a + b$$

fast pointer 이동할 때 : $\text{slow} \times 2$

$$\therefore \text{fast} = 2(a + b)$$

두 포인터가 하나의 위치에서 만날 때

즉, fast 가 slow 두 배 $a+b$ 만큼 더 이동할